

Definite integrals – 1

Darboux sums and criteria for integrability

8 сентября 2026 года

Theoretical minimum.

Def. For a bounded f on $[a, b]$ and a partition P put $L(f, P) = \sum m_i \Delta x_i$ and $U(f, P) = \sum M_i \Delta x_i$, where m_i and M_i are the infimum and the supremum of f on the i -th subinterval. The lower and upper integrals are $\sup_P L$ and $\inf_P U$; f is integrable when they coincide. The *oscillation* of f on a set is $\omega(f, E) = \sup_E f - \inf_E f$.

Theorem 1 (Darboux's criterion). A bounded f is integrable on $[a, b]$ if and only if for every $\varepsilon > 0$ there is a partition with $U(f, P) - L(f, P) < \varepsilon$, equivalently $\sum_i \omega(f, [x_{i-1}, x_i]) \Delta x_i < \varepsilon$. Refining a partition never increases U and never decreases L .

Theorem 2 (which functions are integrable). Continuous functions and monotone functions are integrable; integrability is preserved by sums, products, $|f|$, max and min, and by composing on the outside with a continuous function. Changing f at finitely many points changes neither the integrability nor the value.

Fact 1 (Riemann's form of the criterion). A bounded f is integrable if and only if for every $\varepsilon > 0$ and every $\sigma > 0$ the set $\{x : \omega(f, x) \geq \sigma\}$ can be covered by finitely many intervals of total length less than ε . This is the elementary ancestor of Lebesgue's criterion.

-
1. Prove Theorem 1 in both forms, prove that refinement is monotone, and deduce that continuous and monotone functions are integrable. Then prove that the lower integral never exceeds the upper one for an arbitrary bounded function.

2. Prove that fg , $|f|$ and $\max(f, g)$ are integrable whenever f and g are, and that $\varphi \circ f$ is integrable for continuous φ . Give integrable f and g for which $f \circ g$ is not integrable.
3. (Iranian competition) Let $f: [0, 1] \rightarrow \mathbb{R}$ be bounded and suppose that the left-hand limit of f exists at every point of $[0, 1]$. Prove that f is Riemann integrable.
4. (RMO) Let f be integrable on $[a, b]$ and let $a_n \rightarrow 0$ with $a_n > 0$. Suppose that $|\int_x^y f| \leq |x - y|$ for all $x, y \in [a, b]$ with $|x - y| = a_n$ for some n . Prove that the same inequality holds for all $x, y \in [a, b]$.
5. (RMO) Let $f, g: [a, b] \rightarrow \mathbb{R}$ be such that f has antiderivatives and is bounded above, and $g(y) - g(x) \geq (y - x) \sup_{t \in [x, y]} f(t)$ for all $x < y$. Prove that g is integrable.